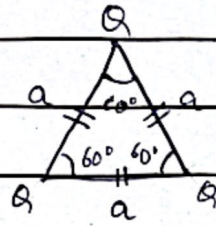


Homework 1

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Q.1)



using the super position formula:

$$F = \frac{Q}{4\pi\epsilon_0} \sum_{k=1}^N \frac{Q_k (r - r_k)}{|r - r_k|^3}$$

$$N = 2$$

$$\therefore F = \frac{Q}{4\pi\epsilon_0} \sum_{k=1}^2 \frac{Q_k (r - r_k)}{|r - r_k|^3}$$

$$= \frac{Q}{4\pi\epsilon_0} \left[\frac{Q(r - r_1)}{|r - r_1|^3} + \frac{Q(r - r_2)}{|r - r_2|^3} \right]$$

$$\text{Since } |r - r_1| = |r - r_2| = a$$

$$F = \frac{Q}{4\pi\epsilon_0} \left[\frac{Q(r - r_1)}{a^3} + \frac{Q(r - r_2)}{|r - r_2|} \right]$$

$$= \frac{Q^2}{4\pi\epsilon_0} \left[\frac{(r - r_1)}{a^3} + \frac{(r - r_2)}{a^3} \right]$$

$$= \frac{Q^2}{4\pi\epsilon_0 a^3} \left[\frac{r - r_1}{|r - r_1|} + \frac{r - r_2}{|r - r_2|} \right]$$

$$\vec{u}_1 = \frac{r-r_1}{|r-r_1|}, \quad \vec{u}_2 = \frac{r-r_2}{|r-r_2|},$$

Hence,

$$F = \frac{Q^2}{4\pi\epsilon_0 a^3} (\vec{u}_1 + \vec{u}_2)$$

angle b/w $\vec{u}_1$ and $\vec{u}_2 = 60^\circ$.

adding $\vec{u}_1$ and $\vec{u}_2$:

$$|\vec{u}_1 + \vec{u}_2| = \sqrt{1^2 + 1^2 + 2\cos 60} = \sqrt{3}$$

$$|r| = \frac{\sqrt{3} Q^2}{4\pi\epsilon_0 a^2}$$

Q2.) $k = \frac{1}{4\pi\epsilon_0}$

a.) $\phi(0) = 0$

$$d\phi = \frac{k dq}{R}$$

$$\phi = \frac{k}{R} \int dq = \frac{k Q_1}{R}$$

$$\phi_R = \frac{k Q_2}{R}, \quad \phi_{2R} = \frac{k Q_2}{2R}$$

$$\phi(0) = \frac{k}{R} \left(Q_1 + Q_2 + \frac{Q_2}{R} \right)$$

$$\phi(0) = \frac{k}{R} \left(Q_1 + \frac{3}{2} Q_2 \right)$$

$$\phi(0) = 0$$

$$\therefore Q_1 + \frac{3}{2} Q_2 = 0$$

$$Q_2 = -\frac{2}{3} Q_1$$

$$b) \quad r' = (R \cos \theta, R \sin \theta)$$

$$dq = \lambda R d\theta$$

field at origin due to dq :

$$dE = \frac{k dq (0 - r')}{|r'|^3}$$

$$= \frac{k dq}{R^3} (-R \cos \theta a_x - R \sin \theta a_y)$$

$$= \frac{k dq}{R^2} (-\cos \theta a_x - \sin \theta a_y)$$

$$dE_y = \frac{k dy}{R^2} (-\sin \theta) = \frac{k \lambda}{R} (-\sin \theta) \cdot d\theta$$

$$E_y, \text{ arc} = \int_{\theta = \pi/4}^{3\pi/2} \frac{k \lambda}{R} (-\sin \theta) \cdot d\theta$$

$$= \frac{k \lambda}{R} [\cos \theta]_{\pi/4}^{3\pi/2} = \frac{k \lambda}{R} (0 - 0) = 0$$

$$dE_x = \frac{k dq}{R^2} (-\cos \theta)$$

$$= \frac{k \lambda}{R} (-\cos \theta) d\theta$$

$$E_{x, arc} = \int_{\theta=\pi/2}^{3\pi/2} \frac{k \lambda}{R} (-\cos \theta) \cdot d\theta$$

$$= \frac{k \lambda}{R} \left[-\sin \theta \right]_{\pi/2}^{3\pi/2}$$

$$= \frac{k \lambda}{R} \left[-(-1) - (-1) \right] = \frac{k \lambda}{R} (2) = \frac{2k \lambda}{R}$$

$$\lambda = \frac{Q_1}{\pi R} \quad \therefore E_{x, arc} = \frac{2k Q_1}{\pi R^2}$$

yield of the two point charges at the centre:

At point charge Q_2 at $x = d > 0$

$$E_d = k Q_2 \frac{0 - r_d}{|0 - r_d|^3}$$

$$= \frac{k Q_2}{d^2} a_x$$

$$d = R \text{ and } d = 2R$$

$$E_{x, R} = \frac{-k Q_2}{R^2}, \quad E_{x, 2R} = \frac{-k Q_2}{4R^2}$$

$$E_{x, R} + E_{x, 2R} = \frac{-5k Q_2}{4R^2}$$

$$\epsilon_{x, \text{arc}} + \epsilon_{x, \text{points}} = 0$$

$$\frac{2kQ_1}{\pi R^2} + \left(\frac{-5kQ_2}{4R^2} \right) = 0$$

$$\frac{2kQ_1}{\pi R^2} = \frac{5kQ_2}{4R^2}$$

$$\therefore Q_2 = \frac{8}{5\pi} Q_1$$

Q.3) $a \leq r \leq b$

Volume charge density $\rho(r) = \frac{K}{r^2}$ ($K = \text{constant}$)

$$K = \frac{1}{4\pi\epsilon_0}, \quad V(\infty) = 0$$

a) Total charge in the shell:

$$Q_{\text{total}} = \int_V \rho(r) \cdot dV$$

$$dV = 4\pi r^2 \cdot dr$$

$$Q_{\text{total}} = \int_a^b \rho(r) 4\pi r^2 \cdot dr$$

$$= \int_a^b \frac{K}{r^2} 4\pi r^2 \cdot dr$$

$$= 4\pi K \int_a^b dr = 4\pi K [r]_a^b$$

$$= 4\pi K (b - a)$$

b) Electric field $E(r)$

$$E(r) = E(r) ar$$

$$E(r) 4\pi r^2 = \frac{Q_{enc}(r)}{\epsilon_0} \quad (\text{Gauss' Law with a sphere of radius } r)$$

Region 1: $0 \leq r \leq a$, No charge enclosed.

$$\therefore Q_{enc} = 0$$

$$\therefore E(r) = 0.$$

Region 2: $a \leq r \leq b$

$$Q_{enc}(r) = \int_a^r \frac{k}{r'^2} 4\pi r'^2 dr'$$

$$= 4\pi k(r-a)$$

$$E(r) = \frac{Q_{enc}}{4\pi \epsilon_0 r^2} = \frac{k}{\epsilon_0} \left(\frac{1}{r} - \frac{a}{r^2} \right)$$

Region 3: $r \geq b$

$$Q_{enc} = Q_{tot} = 4\pi k(b-a)$$

$$E(r) = \frac{Q_{tot}}{4\pi \epsilon_0 r^2} = \frac{k(b-a)}{\epsilon_0 r^2}$$

c) Electric Potential $V(r)$ and $V(\infty) = 0$

$$V(r) = \int_r^\infty E(r') dr'$$

$$V(r) = \int_r^\infty \frac{k(b-a)}{\epsilon_0 r'^2} \cdot dr'$$

$$V(r) = \frac{k(b-a)}{\epsilon_0 r}$$

$$\underline{a \leq r \leq b:}$$

$$V(r) = \int_r^b \frac{k}{\epsilon_0} \left(\frac{1}{r'} - \frac{a}{r'^2} \right) dr' + \int_b^\infty \frac{k(b-a)}{\epsilon_0 r'^2} \cdot dr'$$

$$\int_r^b \left(\frac{1}{r'} - \frac{a}{r'^2} \right) dr' = \left[\ln(r') + \frac{a}{r'} \right]_r^b$$

$$= \ln b - \ln r + a \left(\frac{1}{b} - \frac{1}{r} \right)$$

$$\int_b^\infty \frac{(b-a)}{r'^2} dr' = \frac{b-a}{b}$$

$$V(r) = \frac{k}{\epsilon_0} \left[\ln\left(\frac{b}{r}\right) + a \left(\frac{1}{b} - \frac{1}{r} \right) + \frac{b-a}{b} \right]$$

$$= \frac{k}{\epsilon_0} \left(\ln\left(\frac{b}{r}\right) + 1 - \frac{a}{r} \right)$$

$$\underline{0 \leq r \leq a:}$$

$$E=0 \quad \therefore$$

$$V(a) = \frac{k}{\epsilon_0} \left(\ln\left(\frac{b}{a}\right) + 1 - \frac{a}{a} \right)$$

$$= \frac{k}{\epsilon_0} \left(\ln\left(\frac{b}{a}\right) \right)$$

$$V(r) = \frac{k}{\epsilon_0} \ln\left(\frac{b}{a}\right)$$

d) limit $b \rightarrow a$

$$Q_{\text{tot}} = 4\pi k (b - a) \rightarrow 0$$

$$E(r) \rightarrow 0 \text{ everywhere}$$

$$V(r) \rightarrow 0 \text{ everywhere since } \ln\left(\frac{b}{a}\right) \rightarrow 0.$$

[zero net charge].

Q.4) Consider a ring in the circular disk, where;

Radius of circular disk = a ;

Radius of ring = dr ;

charge on ring = dq .

$$dq = \rho_s ds \quad \therefore ds = 2\pi r dr$$

$$dq = \rho_s 2\pi r dr \quad \text{--- (1)}$$

As we know,

$$E \text{ due to a ring of charge} = \frac{dq h Q}{4\pi\epsilon_0 (b^2 + h^2)^{3/2}} \quad \text{--- (2)}$$

b = distance from origin to ring

h = distance from origin to point P .

Q = charge in ring

$\therefore$ Since

$$b \rightarrow r, \quad h \rightarrow h, \quad Q \rightarrow dq$$

$$\text{Eq. (2) becomes: } dE = \frac{h dq a_y}{4\pi\epsilon_0 (r^2 + h^2)^{3/2}} = \frac{h \rho_s 2\pi r dr a_y}{4\pi\epsilon_0 (r^2 + h^2)^{3/2}}$$

$$\rho_s = C \rho^3$$

$$dE = \frac{h c \rho^3 r dr}{2 \epsilon_0 (r^2 + h^2)^{3/2}} a_y$$

⇒ Integrate from 0 to a, circular disk is sum of all rings of width dr .

$$\int dE = \int_0^a \frac{h c \rho^3 r dr}{2 \epsilon_0 (r^2 + h^2)^{3/2}} a_y = \frac{h c \rho^3 a_y}{2 \epsilon_0} \int_0^a \frac{r \cdot dr}{(r^2 + h^2)^{3/2}}$$

$$E = \frac{h c \rho^3}{2 \epsilon_0} \cdot \int_0^a \frac{1}{2} \cdot (r^2 + h^2)^{-3/2} \cdot 2r \cdot dr$$

$$E = \frac{h c \rho^3}{2 \epsilon_0} \cdot \frac{1}{2} \cdot \frac{(r^2 + h^2)^{-1/2}}{-1/2} \Big|_0^a$$

$$E = \frac{h c \rho^3}{2 \epsilon_0} \cdot (-1) \cdot \left[\frac{1}{(a^2 + h^2)^{1/2}} - \frac{1}{(0^2 + h^2)^{1/2}} \right]$$

$$E = \frac{c \rho^3}{2 \epsilon_0} \cdot \left[\frac{-h}{(a^2 + h^2)^{1/2}} + \frac{h}{(h^2)^{1/2}} \right]$$

$$E = \frac{c \rho^3}{2 \epsilon_0} \left[1 - \frac{h}{\sqrt{a^2 + h^2}} \right] \quad \text{Ans}$$

Q.5) Electric field in spherical coordinate is,

$$E = \frac{c}{r^4} \hat{a}_r \quad (\text{V/m}).$$

$r \rightarrow$ distance from origin to A or B, both A & B are on x-plane, $y=0$ & $z=0$, such that $|x|=r$, $dr=dx$.

$\therefore$ We have,

$$E_x = \pm \frac{20}{x^4} \quad \text{and} \quad E = \frac{c}{r^4} \hat{a}_r$$

$$c = 20$$

We know that $E = -\nabla V$

$$E = - \left[\frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k} \right]$$

$$E_x = - \frac{\partial V}{\partial x} = - \frac{dV}{dx}$$

$$V = - \int E_x \cdot dx$$

We need to find,

$$V_A - V_B = - \int_B^A E_x \cdot dx$$

$$\Rightarrow V_{AB} = - \int_{-4}^2 \frac{20}{x^4} \cdot dx = -20 \int_{-4}^2 (x)^{-4} \cdot dx$$

$$= -20 \cdot \frac{x^{-3}}{-3} \Big|_{-4}^2 = \frac{20}{3} \left[\frac{1}{2^3} - \frac{1}{(-4)^3} \right] = \frac{15}{16} \text{ V}$$

$$Q.6) \rho_v = c \rho e^{-\rho}, \quad c > 0$$

$$D = D_\rho(\rho) a_\rho$$

$$a) \oint_S D \cdot ds = Q_{enc.} -$$

$$\oint_S D \cdot ds = D_\rho(\rho) (2\pi \rho L)$$

$$Q_{enc} = \int_V \rho_v \cdot dV \quad \therefore dV = \rho' d\rho' d\phi dz$$

$$Q_{enc} = \int_0^L \int_0^{2\pi} \int_0^\rho c \rho' e^{-\rho'} (\rho' d\rho' d\phi dz)$$

$$= \rho'^2 e^{-\rho'} c \int_0^\rho \int_0^{2\pi} \int_0^L dz d\phi d\rho'$$

$$= \rho'^2 e^{-\rho'} c \int_0^\rho \int_0^{2\pi} L \cdot d\phi d\rho'$$

$$= \rho'^2 e^{-\rho'} c \int_0^\rho 2\pi L d\rho'$$

$$= c (2\pi L) \int_0^\rho \rho'^2 e^{-\rho'} d\rho'$$

$$D_\rho(2\pi \rho L) = c(2\pi L) \int_0^\rho \rho' e^{-\rho'} \cdot d\rho'$$

$$D_\rho(\rho) = \frac{c}{\rho} \int_0^\rho \rho'^2 e^{-\rho'} d\rho'$$

$$\int_0^\rho \rho'^2 e^{-\rho'} d\rho' = 2 - e^{-\rho} (\rho^2 + 2\rho + 2)$$

$$D_\rho(\rho) = \frac{c}{\rho} [2 - e^{-\rho} (\rho^2 + 2\rho + 2)]$$

$$D = D_\rho(\rho) a_\rho = \frac{c}{\rho} [2 - e^{-\rho} (\rho^2 + 2\rho + 2)] a_\rho$$

$$b.) \nabla \cdot D = \rho_v$$

$$\nabla \cdot D = \frac{1}{\rho} \cdot \frac{d}{d\rho} (\rho D_\rho)$$

$$\Rightarrow \frac{1}{\rho} \cdot \frac{d}{d\rho} (\rho D_\rho) = c \rho e^{-\rho}$$

$$\Rightarrow \frac{d}{d\rho} (\rho D_\rho) = c \rho^2 e^{-\rho}$$

$$\Rightarrow \rho D_\rho = c \int_0^\rho \rho' e^{-\rho'} d\rho'$$

$$\Rightarrow D_\rho = \frac{c}{\rho} [2 - e^{-\rho} (\rho^2 + 2\rho + 2)]$$

$$\Rightarrow D = \frac{c}{\rho} [2 - e^{-\rho} (\rho^2 + 2\rho + 2)] a_\rho$$

$$Q.7) E(r) = \int_L \frac{\rho_L dl' (r-r')}{4\pi\epsilon_0 |r-r'|^3}$$

$$r=0, r-r' = -r' = -a \cos \phi a_x - a \sin \phi a_y$$

$$\frac{r-r'}{|r-r'|^3} = \frac{-r'}{|r'|^3} = \frac{-1}{a^3} (a \cos \phi a_x + a \sin \phi a_y)$$

$$= \frac{-1}{a^2} (\cos \phi a_x + \sin \phi a_y)$$

$$E(0) = \int_0^\pi \frac{\rho_L}{4\pi\epsilon_0} \left[\frac{-1}{a^2} (\cos \phi a_x + \sin \phi a_y) \right] (a d\phi)$$

$$= -\frac{\rho_L}{4\pi\epsilon_0} \int_0^\pi (\cos \phi a_x + \sin \phi a_y) d\phi$$

$$\int_0^\pi \cos \phi d\phi = 0, \int_0^\pi \sin \phi d\phi = 2.$$

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$$E(0) = \frac{-\rho_L}{4\pi\epsilon_0} (0a_x + 2a_y) = \boxed{\frac{-\rho_L}{2\pi\epsilon_0} a_y}$$